

Owen Arden

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<https://owenarden.github.io/home/>

EMPLOYMENT HISTORY

2024–Present	Associate Professor of Computer Science University of California Santa Cruz, Santa Cruz, CA
2017–2024	Assistant Professor of Computer Science University of California Santa Cruz, Santa Cruz, CA
2020–2021	Visiting Scientist Facebook, Inc. (On leave from UCSC)
2016–2017	Postdoctoral Fellow Harvard University, Cambridge, MA (On leave from UCSC)
2009–2010	Research Adjunct Institute for Defense Analyses Center for Computing Sciences, Bowie, MD
2008–2009	Security Analyst Independent Security Evaluators, Baltimore, MD
2004–2008	Global Network Vulnerability Analyst National Security Agency, Fort Meade, MD

EDUCATION

2009–2017	Ph.D., Computer Science Cornell University, Ithaca, NY
1999–2003	B.S. (Highest Honors), Computer Engineering Georgia Institute of Technology, Atlanta, GA

SELECTED HONORS AND AWARDS

- [H6] **Distinguished Artifact Award.** *IEEE Dependable Systems and Networks* 2024.
- [H5] **Distinguished Paper Award.** *IEEE Computer Security Foundations Symposium* 2022.
- [H4] **Member.** Information Sci. and Tech. (ISAT) Study Group, DARPA. 2020-2025
- [H3] **Best Paper Finalist.** *ACM Conf. on Computer and Communications Security* 2017.
11 finalists selected from 151 accepted papers.
- [H2] **Best Paper.** *Biennial Conference on Innovative Data Systems Research* 2013.
- [H1] **National Intelligence Meritorious Unit Citation** (NIMUC). Office of the Director of National Intelligence (ODNI), 2007. The NIMUC is the 7th highest award presented by the ODNI across all US intelligence agencies.

GRANTS

- [G3] Principal Investigator, Berkeley RDI AI & Decentralization Innovation Award. *Approximate but accurate: Improving robustness to Byzantine manipulation in multi-agent systems.*, Spring 2025. \$15,000.

- [G2] Co-Principal Investigator, IARPA HECTOR Project, *PASCAL: Programming Architecture for Secure Cryptographic AppLications*, 2019-2024, \$15,477,510 (Original direct award: \$864,699; Distributed: \$147,531).
- [G1] Principal Investigator, National Science Foundation (NSF CAREER Award), *Building Secure Decentralized Applications with Trusted Hardware and Blockchains*, 2018-2023, \$510,767.

SCHOLARLY AND CREATIVE WORK

JOURNAL ARTICLES

- [J2] Priyanka Mondal, Maximilian Algehed, and Owen Arden. “Flow-Limited Authorization for Consensus, Replication, and Secret Sharing”. In: *Journal of Computer Security* 31.5 (Oct. 2023). Ed. by Stefano Calzavara and David Naumann, pp. 615–645
- [J1] Jed Liu, Owen Arden, Michael D. George, and Andrew C. Myers. “Fabric: Building Open Distributed Systems Securely by Construction”. In: *J. Computer Security*. JCS’17 25.4–5 (May 2017), pp. 319–321

CONFERENCE ARTICLES

- [C19] Roy Shadmon and Owen Arden. “Enhancing Accuracy in Approximate Byzantine Agreement with Bayesian Inference”. In: *2025 55th Annual IEEE/IFIP International Conference on Dependable Systems and Networks - Supplemental Volume (DSN-S)*. Naples, Italy: IEEE, June 2025, pp. 191–195
- [C18] Haofan Zheng, Tuan Tran, Roy Shadmon, and Owen Arden. “Decentagram: Highly-available Decentralized Publish/Subscribe Systems”. In: *2024 54th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*. DSN’24. 2024
Note: Distinguished artifact award.
- [C17] Tuan Tran, Faisal Nawab, Peter Alvaro, and Owen Arden. “Unstick Yourself: Recoverable Byzantine Fault Tolerant Services”. In: *2023 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*. ICBC’23. May 2023
- [C16] Priyanka Mondal, Maximilian Algehed, and Owen Arden. “Applying Consensus and Replication Securely with FLAQR”. in: *35th IEEE Computer Security Foundations Symp. (CSF)*. CSF’22. Aug. 2022 **Note: Distinguished paper award.**
- [C15] Tuan Tran, Haofan Zheng, Peter Alvaro, and Owen Arden. “Payment Channels under Network Congestion”. In: *2022 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*. ICBC’22 (Short Paper). May 2022
- [C14] John H. Castellanos, Martin Ochoa, Alvaro A. Cardenas, Owen Arden, and Jianying Zhou. “AttckFinder: Discovering Attack Vectors in PLC Programs Using Information Flow Analysis”. In: *24th International Symposium on Research in Attacks, Intrusions and Defenses*. RAID’21. 2021, pp. 235–250
- [C13] Haofan Zheng and Owen Arden. “Secure Distributed Applications the Decent Way”. In: *Proceedings of the 2021 International Symposium on Advanced Security on Software and Systems*. ASSS’21. Jan. 2021, pp. 29–42
- [C12] Haofan Zheng, Tuan Tran, and Owen Arden. “Total Eclipse of the Enclave: Detecting Eclipse Attacks from inside Tees”. In: *2021 IEEE International Conference on Blockchain and Cryptocurrency (ICBC)*. ICBC’21 (Short Paper). May 2021
- [C11] Hirsch, Andrew K., Azevedo de Amorim, Pedro Henrique, Cecchetti, Ethan, Tate, Ross, and Arden, Owen. “First-Order Logic for Flow-Limited Authorization”. In: *33rd IEEE Computer Security Foundations Symp. (CSF)*. CSF’20. 2020, pp. 123–138

- [C10] Daniel Abadi, Owen Arden, Faisal Nawab, and Moshe Shadmon. “AnyLog: A Grand Unification of the Internet of Things”. In: *Conference on Innovative Data Systems Research (CIDR)*. CIDR’20. 2020
- [C9] Anitha Gollamudi, Stephen Chong, and Owen Arden. “Information Flow Control for Distributed Trusted Execution Environments”. In: *32nd IEEE Computer Security Foundations Symp. (CSF)*. CSF’19. June 2019
- [C8] Ethan Cecchetti, Andrew C. Myers, and Owen Arden. “Nonmalleable Information Flow Control”. In: *24th ACM Conf. on Computer and Communications Security (CCS)*. CCS’17. Dallas, TX, Oct. 2017, pp. 1875–1891 **Note: Best paper finalist (11 finalists out of 151 accepted papers)**.
- [C7] Lucas Waye, Pablo Buiras, Owen Arden, Alejandro Russo, and Stephen Chong. “Cryptographically Secure Information Flow Control on Key-Value Stores”. In: *24th ACM Conf. on Computer and Communications Security (CCS)*. CCS’17. 2017, pp. 1893–1907
- [C6] Owen Arden and Andrew C. Myers. “A Calculus for Flow-Limited Authorization”. In: *2016 IEEE 29th Computer Security Foundations Symposium (CSF)*. CSF’16. June 2016
- [C5] Owen Arden, Jed Liu, and Andrew C. Myers. “Flow-Limited Authorization”. In: *28th IEEE Computer Security Foundations Symp. (CSF)*. July 2015, pp. 569–583
- [C4] Jed Liu, Tom Magrino, Owen Arden, Michael D. George, and Andrew C. Myers. “Warranties for Faster Strong Consistency”. In: *11th USENIX Symp. on Networked Systems Design and Implementation (NSDI)*. NSDI’14. Apr. 2014, pp. 513–517
- [C3] Alvin Cheung, Samuel Madden, Armando Solar-Lezama, Owen Arden, and Andrew C. Myers. “StatusQuo: Making Familiar Abstractions Perform Using Program Analysis”. In: *Conference on Innovative Data Systems Research (CIDR)*. CIDR’13. Jan. 2013 **Note: Best paper award**.
- [C2] Alvin Cheung, Samuel Madden, Owen Arden, and Andrew C. Myers. “Automatic Partitioning of Database Applications”. In: vol. 5. *VLDB’12*. Aug. 2012, pp. 1471–1482
- [C1] Owen Arden, Michael D. George, Jed Liu, K. Vikram, Aslan Askarov, and Andrew C. Myers. “Sharing Mobile Code Securely with Information Flow Control”. In: *IEEE Symp. on Security and Privacy*. SSP’12. May 2012, pp. 191–205

MAGAZINE ARTICLES

- [M1] Alvin Cheung, Samuel Madden, Armando Solar-Lezama, Owen Arden, and Andrew C. Myers. “Using Program Analysis to Improve Database Applications”. In: *IEEE Data Bulletin* 37.1 (Mar. 2014), pp. 48–59

PATENTS

- [P1] Moshe Shadmon, Levy Cohen, Daniel Abadi, and Owen Arden. *System and apparatus to manage data using a peer-to-peer network and the blockchain*. US Patent 10,868,865. Dec. 2020

PROFESSIONAL ACTIVITIES

PROGRAM COMMITTEES

2026 Architectural Support for Programming Languages and Operating Systems (ASPLOS)
2026 Computer Security Foundations Symposium (CSF)
2025 (General chair) Computer Security Foundations Symposium (CSF)
2025 Architectural Support for Programming Languages and Operating Systems (ASPLOS)
2024 Computer Security Foundations Symposium (CSF)
2024 Choreographic Programming Workshop @ PLDI
2024 Programming Language Design and Implementation (PLDI)
2023 Conference on Computer and Communications Security (CCS)
2023 Computer Security Foundations Symposium (CSF)
2022 Conference on Computer and Communications Security (CCS)
2022 Workshop on Principles of Secure Compilation (PriSC) @ POPL
2022 Computer Security Foundations Symposium (CSF)
2021 Workshop on Programming Languages and Analysis for Security (PLAS), @ CCS
2021 (Co-chair) Workshop on Foundations of Computer Security (FCS), @ CSF
2021 Computer Security Foundations Symposium
2020 Workshop on Programming Languages and Analysis for Security (PLAS), @ CCS
2020 (Co-chair) Symposium on Foundations and Applications of Blockchain (FAB)
2020 (Co-chair) Workshop on Foundations of Computer Security (FCS), @ CSF
2019 (Chair) Workshop on Hardware Security, @ CROSS
2019 Conference on Computer and Communications Security (CCS)
2019 Workshop on Programming Languages and Operating Systems, @ SOSP
2019 Conference on Principles of Security and Trust (POST)
2019 Workshop on Foundations of Computer Security, @ CSF
2018 Workshop on Programming Languages and Analysis for Security (PLAS), @ CCS
2018 Computer Security Foundations Symposium (CSF)
2018 Workshop on Foundations of Computer Security (FCS), @ CSF
2017 Workshop on Foundations of Computer Security (FCS), @ CSF

EXTERNAL REVIEW AND ARTIFACT EVALUATION COMMITTEES

2023 OOPSLA'23 External Review / Artifact Evaluation Committee
2018 USENIX Annual Technical Conference (ATC) External Review Committee
2016 Programming Language Design and Implementation (PLDI) External Review Committee

MISCELLANEOUS SERVICE AND CONSULTING

2023 Judge, Congressional App Challenge (for Rep. Jimmy Panetta)
2020-2025 Member, Information Science and Technology (ISAT) Study Group, DARPA
Summer 2023 AirBnB (Visiting Scientist, Viaduct project)
Winter 2020 Facebook (Privacy Advisory Committee)
2018–2020 AnyLog Inc. (Security Architect)

UNIVERSITY SERVICE

2024–2025	CSE Undergraduate Director
2024–2025	CSE Colloquium Committee
2023–2024	CSE Student Success Committee
2023–2024	CSE Faculty Search Committee
2022–2023	CSE Personnel Committee
2021–2022	CSE Space Committee
2021–2023	UCSC Committee on Career Advising (CCA)
2018–2020	CSE Junior Faculty Lunch Seminar (co-organizer)
2018–2020	CSE Faculty Search Committee
2017–2019	BSOE Graduate Awards Committee
2017–2018	CS Graduate Committee

MENTORING AND STUDENT ADVISING

PHD STUDENTS

2023–Present	Luis Burbano	(Ph.D. committee)
2023–Present	Gan Shen	(Ph.D. committee)
2022–2025	Keerthi Koneru	(Ph.D. committee)
2021–2023	Sohum Banerjea	(Ph.D. committee)
2019–Present	Roy Shadmon	(Primary advisor)
2018–2024	Tuan Tran	(Co-advisor) <i>now at Nutanix</i>
2017–2024	Priyanka Mondal	(Primary advisor / Co-advisor) <i>now at Snap</i>
2017–2024	Haofan Zheng	(Primary advisor) <i>now at TikTok</i>
2018–2019	Thomas Schmitz	(Ph.D. committee)

MASTERS STUDENTS

Winter-Spring 2025	Sulayman Asif	(MS Project chair)
Spring 2025	Ismail Ahmed	(MS Project reader)
Spring 2024	Shun Kashiwa	(MS thesis reader) <i>now PhD at UCSD</i>
Winter 2023	Aakash Mishra	(MS project chair)
Winter 2022	David Lung	(MS project chair)
Fall 2021	Shingo Nakagaki	(MS project reader)
Fall 2020	Bobby Dhillon	(MS project reader)
Spring 2020	Aasim Khan	(MS project reader)
2019–2020	Ana McTaggart	(Research and MS project chair) <i>now at RedHat</i>
Spring 2019	Kuo Pei	(MS project reader)
2018–2019	Roy Shadmon	(Independent study advisor and MS project chair)
Winter–Summer 2018	Christine Quintana	(MS project chair)
Winter–Spring 2018	Kavya Jha	(Independent study advisor)
Winter–Spring 2018	Tanay Parekhji	(Independent study advisor)
Spring-Summer 2020	Jason Vranek	(MS project reader)
Fall 2018	Ziye Han	(MS project chair)
Fall 2018	Shuai Zhou	(MS project chair)

UNDERGRADUATE STUDENTS

Summer 2025	Surendra Jammishetti	(PI, Research Pathways Fellowship)
Winter-Spring 2025	Surendra Jammishetti	(Thesis chair, ind. study)
Winter-Spring 2025	Julian Martinov	(Thesis chair, ind. study)
Spring 2025	Zephiris Evergreen	(Thesis reader)
Winter 2020	Samuel Arnold	(Thesis chair)
Spring 2019	Bobby Dhillon (Thesis reader)	
Spring 2019	Thomas Pham (Thesis reader)	
Spring 2019	Ryan Manseau (Thesis reader)	
Winter–Spring 2018	Manthan Mallikarjun	(Independent study advisor)
Winter–Spring 2018	Akul Goyal	(Independent study advisor)

COURSE RELIEF AND LEAVE OF ABSENCES

Winter 2025	Undergraduate Director CR
Winter 2022	ASMD course relief (family medical leave)
2020-2021	Leave of Absence (Visiting Scientist at Facebook)
Winter 2018	ASMD course relief (parental leave)
Spring 2018	Startup course relief